



Impact of growth conditions and role of *sigB* on *Listeria monocytogenes* fitness in single and mixed biofilms cultured with *Lactobacillus plantarum*

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ABSTRACT

The role of *sigB*, a major transcriptional regulator of stress response genes, was assessed in formation of single and mixed species biofilms of *Listeria monocytogenes* EGD-e and *Lactobacillus plantarum* WCFS1 as secondary species at 20 °C and 30 °C using different medium compositions (nutrient-rich medium with and without supplementary manganese, glucose and salt). *L. monocytogenes* showed significant biofilm development at both temperatures and in all media tested although less biofilm was formed when glucose was supplemented only. The contribution of *L. monocytogenes* to the mixed species biofilm declined especially at higher temperature in glucose-rich medium in the absence and presence of manganese, due to lactic acid formation with concomitant decrease in culture pH below the pH_{min} of *L. monocytogenes*. Using an in-frame *sigB* deletion mutant and a complementation mutant we showed that *sigB* contributed to survival under these acid stress conditions. Notably, the additional presence of salt protected *L. monocytogenes* in the acidic mixed species biofilms resulting in an increase of around 2–3 log₁₀ cfu/ml and this phenomenon showed to be *sigB*-dependent.

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1. Introduction

Biofilms are surface associated structures of bacterial cells that are attached to a substratum, embedded in a matrix of extracellular polymeric substances (Donlan & Costerton, 2002). Bacteria in a biofilm have altered phenotypes compared to cells in liquids with respect to their growth rate and gene transcription, and are in general more resistant to environmental adverse conditions (Renier, Hébraud, & Desvaux, 2011; van der Veen & Abee, 2011). Cells in the biofilm supernatant may also differ from planktonic free-living populations since cell aggregates or clusters detach or are being sloughed from biofilms (Donlan & Costerton, 2002; Stoodley et al., 2001).

In natural environments biofilms are conceivably composed of more than one species resulting in so-called mixed species biofilms (Pereira da Silva & Pereira De Martinis, 2013; Stewart & Franklin, 2008). The ubiquitous foodborne pathogen *Listeria monocytogenes* is able to form biofilms with bacterial species normally present in food environments such as *Lactobacillus* species (Caggia, Randazzo, Di Salvo, Romeo, & Giudici, 2004; Campanini, Pedrazzoni, Barbuti, & Baldini, 1993; van der Veen & Abee, 2011), but limited information is available on how

the relative contributions of *L. monocytogenes* and *Lactobacillus* species in biofilms are influenced by temperature and medium composition. Both *L. monocytogenes* and *Lactobacillus plantarum* are fermentative bacteria but with differences in their pH growth limits (McDonald, Flemming, & Hassan, 1990; Pintado, Raimbault, & Guyot, 2005; van der Veen, Moezelaar, Abee, & Wells-Bennik, 2008). *L. monocytogenes* interaction in a biofilm when co-cultured with species isolated from meat and dairy environments such as *Flavobacterium*, *Staphylococcus sciuri*, *Lactococcus lactis*, *Staphylococcus xylosus* and *Pseudomonas fragi* depends on many interrelated aspects including intrinsic and extrinsic factors (Habimana, Guillier, Kulakauskas, & Briandet, 2011; Habimana, Meyrand, Meylheuc, Kulakauskas, & Briandet, 2009; Jeong & Frank, 1994; Leriche & Carpentier, 2000; Norwood & Gilmour, 2001). So far, little is known about the stress resistance mechanisms of *L. monocytogenes* that contribute to biofilm formation (van der Veen & Abee, 2010, 2011). Sigma B is a general stress response regulator that is activated upon acid, osmotic and oxidative stress conditions but also during *L. monocytogenes* single species biofilm formation (Ferreira, O'Byrne, & Boor, 2001; Utratna, Shaw, Starr, & O'Byrne, 2011; van der Veen & Abee, 2011; van Schaik & Abee, 2005). This points to an important role for Sigma B in cell stress tolerance responses (Abram et al., 2008; Wemekamp-Kamphuis et al., 2004). Knowing the food safety risk of (cross-)adaptive stress responses to environmental stimuli and stress tolerance development, it is still not clear how bacterial interaction contributes to persistence in

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food-processing environments (Bowman, Hages, Nilsson, Kocharunchitt, & Ross, 2012; Lou & Yousef, 1997; O'Driscoll, Gahan, & Hill, 1997; Rajkovic et al., 2009; Ringus, Ivy, Wiedmann, & Boor, 2012). *L. monocytogenes* has been involved in multiple outbreaks and it may be related to the resistance gained when forming biofilms (Czajka & Batt, 1994; Ferreira, Wiedmann, Teixeira, & Stasiewicz, 2014; Majjala et al., 2001). However, to our knowledge only one study explored the role of *sigB* in *L. monocytogenes* mixed species biofilm formation in brain heart infusion medium (BHI) using *Pseudomonas fluorescens* as a secondary organism (Schwab, Hu, Wiedmann, & Boor, 2005). So far no study investigated the role of *sigB* in *L. monocytogenes* biofilm formation as function of temperature and medium composition when cultured as single species and co-cultured with sugar fermenting homolactic *Lactobacillus* spp. producing high acidity depending on availability of sugars and levels of manganese (Archibald, 1986; Carr, Chill, & Maida, 2002).

Therefore, in our study, the role of *sigB* in survival capacity of *L. monocytogenes* EGD-e in single and mixed species biofilms with *L. plantarum* WCFS1 was investigated using a *sigB* deletion mutant and its complemented counterpart. Biofilm formation was assessed in nutrient-rich medium with(out) co-culture with *L. plantarum* WCFS1 at 20 °C and 30 °C. Various components either present in food products or added during processing were considered (manganese, glucose and salt), in order to evaluate their influence on *L. monocytogenes* performance in single and mixed species biofilms.

2. Materials and methods

2.1. Strains, media and growth conditions

L. monocytogenes EGD-e wild-type strain, $\Delta sigB$ mutant strain (EGD-e $\Delta sigB$ with chromosomal in-frame deletion (Chatterjee et al., 2006) and the complementation mutant (*csigB*: EGD-e $\Delta sigB$ with genomic complementation of *sigB* using plasmid pIMK-*sigB* (van der Veen & Abee, 2010)) (Table S1) were stored in BHI broth (Becton Dickinson, Le Pont de Claix, France) containing 15% sterile glycerol (Sigma-ALDRICH, Germany) at –80 °C. *L. plantarum* WCFS1 was stored in De Man, Rogosa and Sharpe (MRS) broth (Merck, Darmstadt, Germany) containing 15% sterile glycerol at –80 °C. BHI or MRS agar plates were streaked with cells from the bacterial stocks using an inoculation needle and plates were incubated at 30 °C for 48 h. Single colonies were inoculated in BHI broth, BHI broth with addition of 0.005% (w/v) manganese sulphate (Merck, Darmstadt, Germany) (BHI-Mn), based on the concentration of the specific *Lactobacillus* medium MRS (de Man, Rogosa, & Sharpe, 1960), BHI broth with 2% (w/v) glucose (Merck, Darmstadt, Germany) (BHI-G), BHI broth with addition of 0.005% manganese sulphate and 2% glucose (BHI-Mn-G), or BHI-Mn-G with 3% of salt (BHI-Mn-G-salt) and grown for 18 h at 30 °C, 200 rpm.

2.2. Biofilm formation

Static biofilms were grown in 12-well polystyrene microtiter plates (22 mm diameter, Greiner Bio-One, Frickenhausen, Germany) following the protocol described by van der Veen and Abee (2011) with minor modifications. Overnight grown cultures were inoculated (1%) in wells containing 3 ml BHI, BHI-Mn, BHI-G, BHI-Mn-G and BHI-Mn-G-salt. The plates were statically incubated at 20 °C and 30 °C for 24 and 48 h, and the biofilm supernatant was removed by pipetting. Supernatants were used for pH measurements and enumeration. The biofilms were washed three times with phosphate buffered saline (PBS) (Merck, Darmstadt, Germany) and dispersed in 1 ml PBS per well by pipetting rigorously. Serial dilutions of the biofilm cells and supernatant were made in PBS and BHI when the pH needed to be neutralised, and appropriate dilutions were plated on BHI agar (for *L. monocytogenes* single biofilms), Agar Listeria according To Ottaviani and Agosti (ALOA) (for *L. monocytogenes* mixed biofilms) and MRS agar (for *L. plantarum*). The agar plates were incubated aerobically or microaerophilically for MRS agar plates for

2 days at 30 °C, and colonies were enumerated. Static biofilm formation levels were determined in three biologically independent experiments.

2.3. HPLC-DAD experiments

The diluted samples (1:10) of the filter sterilised (0.2 µm) supernatant of single and mixed species biofilms obtained after 2 days of incubation were analysed for organic acids by high-performance liquid chromatography with diode-array detection (HPLC-DAD) following the protocol described by Knol, Linssen, and van Boekel (2010).

2.4. Statistical analysis

A Student's test ($\alpha = 0.05$) was used to test the significance of the differences between mean $\log_{10}$ counts of the viable cells in biofilms and supernatant and between pH values obtained at the different conditions.

3. Results and discussion

3.1. Single and mixed species biofilm formation of *L. monocytogenes* EGD-e in co-culture with *L. plantarum* WCFS1

In this study, single and mixed species biofilm and supernatant cells of *L. monocytogenes* EGD-e in co-culture with *L. plantarum* WCFS1 were quantified by plate counts in BHI, BHI supplemented with either Mn (BHI-Mn), glucose (BHI-G) or both (BHI-Mn-G), at 20 °C and 30 °C to investigate the factors that affect the biofilm formation of *L. monocytogenes*. *L. monocytogenes* biofilms formed at 20 °C in BHI, BHI-Mn and BHI-Mn-G reached numbers around 7.5–8 $\log_{10}$ cfu/ml (Fig. 1A) with no significant differences between the media used ($P > 0.05$). However, the addition of only glucose to BHI (BHI-G) resulted in significantly lower biofilm cell counts (5.5 $\log_{10}$ cfu/ml) and supernatant cell counts at 20 °C ($P < 0.05$) that was overcome with the simultaneous supplementation of manganese (BHI-Mn-G). Both for BHI-G and BHI-Mn-G a mild acidic condition was reached (pH 4.8 and pH 4.5, respectively). Previously, it has been shown that the higher carbohydrate consumption (ca. 4 g/l) of *L. monocytogenes* 5873 (1/2a) resulted in lower numbers of adhered cells (Saá, Cabo, & Rodríguez, 2009). In line with this observation, we also observed that the addition of only glucose and not $MnSO_4$ reduced exponential growth of *L. monocytogenes* in the planktonic phase in BHI compared to the other three media (Fig. S1). The pH difference between BHI-G and BHI-Mn-G could not explain the enhanced growth in the manganese supplemented condition and suggest an influence of manganese. Vasconcelos and Deneer (1994) indicated that superoxide dismutase (SOD) in *L. monocytogenes* was most probably Mn-dependent, and Suo, Huang, Liu, Shi, and Shi (2012) demonstrated that SOD played an important role in *L. monocytogenes* 4b G biofilm formation coping with oxidative stress. This might point to a potential role of manganese in *Listeria* biofilm formation. In addition, Hassett et al. (1999) showed that quorum sensing also has a role in controlling the expression of superoxide dismutase and catalase genes in *Pseudomonas aeruginosa*. These studies suggest that cells encounter oxidative stress in the biofilms. Our results suggested that mild acid stress during biofilm formation in glucose-rich medium might contribute to biofilm-associated oxidative stress, as observed for *B. cereus* in aerobically grown acidified cultures (Mols, van Kranenburg, van Melis, Moezelaar, & Abee, 2010). Therefore, manganese might play a role as radical scavenger and/or as co-factor of critical enzymes in oxidative stress management during biofilm development of *L. monocytogenes* EGD-e in glucose-rich environment. While for *L. monocytogenes* the role of manganese in stress management is unexplored, the role of manganese in the fitness of *Lactobacillus plantarum* is well known (Archibald & Fridovich, 1981; Watanabe, van der Veen, Nakajima, & Abee, 2012), and showed to be critical in optimal fermentation and coping with oxidative stress as oxygen radical scavenger.

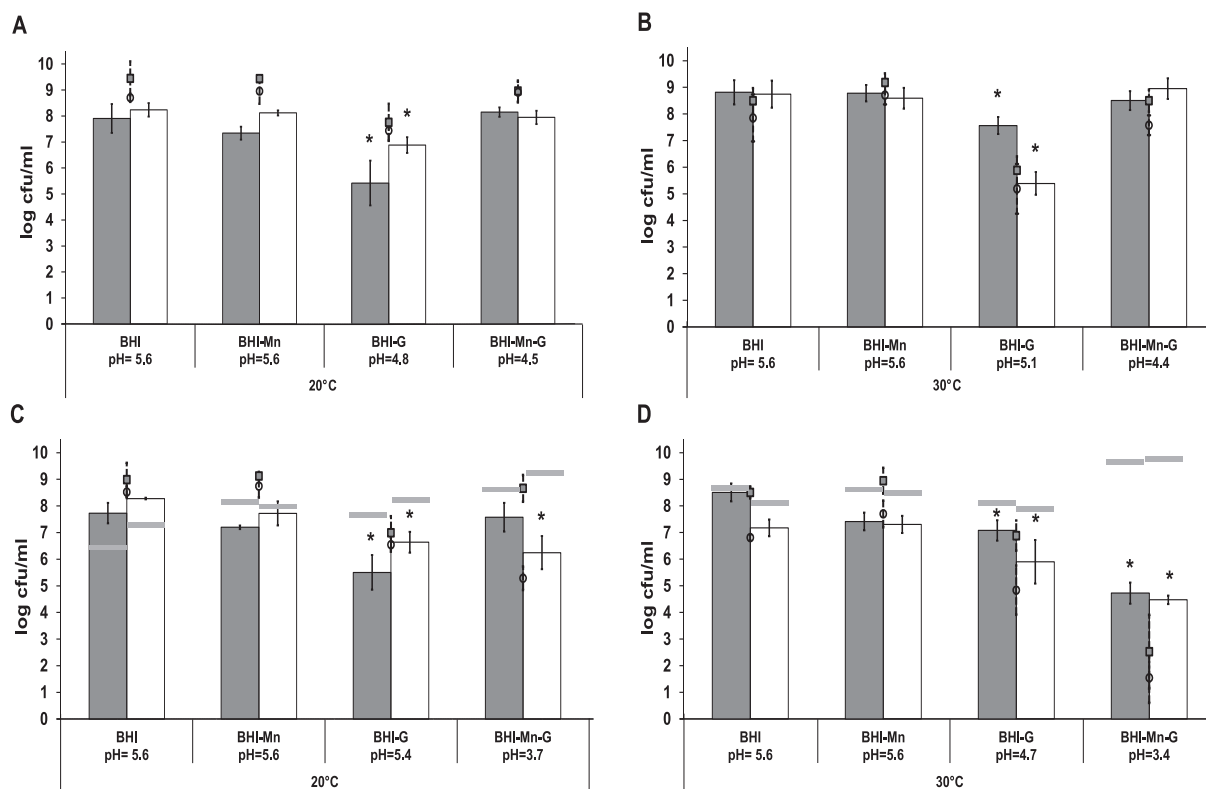


Fig. 1. *L. monocytogenes* EGD-e biofilm and supernatant counts mono-cultured (A, B, top) and co-cultured (C, D, bottom) with *L. plantarum* WCFS1 at 20 °C (A, C, left) and 30 °C (B, D, right) respectively in 12-well polystyrene plates in BHI, BHI-Mn, BHI-G and BHI-Mn-G. The graphs show the average *L. monocytogenes* EGD-e biofilm counts (log₁₀ cfu/ml, bars after 24 h (grey) and 48 h (white)) and *L. monocytogenes* supernatant counts (log₁₀ cfu/ml symbols after 24 h (square) and 48 h (circle)) of 3 biologically independent experiments; *L. plantarum* WCFS1 average biofilm counts (log₁₀ cfu/ml) in the mixed biofilms (C, D: light grey lines after 24 h and 48 h, supernatant counts not shown). *: biofilm levels significantly different from BHI. pH measured after 48 h.

Also at 30 °C the biofilm counts were lower in BHI-G compared to the other media (Fig. 1B), although significant faster biofilm formation was observed for the four tested media at 30 °C. Previous studies also observed that the adherence of *L. monocytogenes* and biofilm development were better at higher temperatures, closer to its optimum growth temperature (Kadam et al., 2013; Moltz, 2005).

The contributions of *L. monocytogenes* and *L. plantarum* in the dual species biofilms were affected by the addition of glucose and/or manganese sulphate to BHI (Fig. 1C). When co-cultured with *L. plantarum* at 20 °C, the concentration of *L. monocytogenes* in the biofilm formed in BHI, BHI-Mn and BHI-G (Fig. 1C) was comparable to that obtained in single species biofilm (Fig. 1A) ($P > 0.05$). However, the biofilm levels of *L. monocytogenes* in BHI-Mn-G declined significantly when co-cultured with *L. plantarum* ($P < 0.05$), which is in agreement with previous results at 20 °C (van der Veen & Abee, 2011). When *L. plantarum* reached high levels in BHI-Mn-G at 48 h the difference between both species was about 3 log₁₀ cfu/ml. The temperature also affected the bacterial species predominance in the mixed species biofilm. In BHI the contribution of *L. plantarum* to the mixed biofilm was favoured by a growth temperature of 30 °C (Fig. 1D) over 20 °C (Fig. 1C). This is reflected by the 1 log₁₀ cfu/ml lower *L. plantarum* numbers at 20 °C compared to *L. monocytogenes*. The *L. plantarum* contribution significantly increased at 30 °C (8.7 log₁₀ cfu/ml after 24 h) exceeding the partner species after 48 h (~1 log₁₀ cfu/ml) due to a reduction in *L. monocytogenes* cell counts. Interestingly, in BHI-Mn-G at 30 °C the numbers of *L. plantarum* were the highest reaching close to 10 log₁₀ cfu/ml; on the contrary were the lowest for *L. monocytogenes* (~4.5 log₁₀ cfu/ml), especially for the supernatant cells (~1.5 log₁₀ cfu/ml), whereas *L. plantarum* supernatant cells remained around ~8 log₁₀ cfu/ml (data not shown). Planktonic monocultured *L. monocytogenes* strains cannot grow in BHI at pH

5.2 or lower when exposed to NaLac (van der Veen et al., 2008), whereas *L. plantarum* WSO resists acidification down to pH 3.0 in modified HHD medium with added lactate and glucose at 30 °C (McDonald et al., 1990). In our study, the addition of MnSO₄ to BHI-G allowed fast acidification of the medium by *L. plantarum* to a pH below the growth limit for *L. monocytogenes*. At 30 °C after 48 h, the glucose was completely fermented into lactic acid only, as no other organic acids were detected in HPLC-DAD experiments. Several studies reported the antilisterial effect of lactic acid bacteria in different food products (Aguilar, Vanegas, & Klotz, 2011; Amézquita & Brashears, 2002; Héquet et al., 2007; Mills et al., 2011; Tomé, Gibbs, & Teixeira, 2008; Vermeiren, Devlieghere, Vandekinderen, & Debevere, 2006), but remarkably, *L. monocytogenes* could survive for a long time under these stressful conditions within the mixed biofilm structure because even after 6 days, 15 colonies were isolated from the mixed biofilm at 30 °C whereas no viable *L. monocytogenes* cells were detected in the supernatant from day 3 onwards. The difference between *L. monocytogenes* counts in the supernatant and in the biofilm underlined the protective biofilm mode. *L. monocytogenes* acid tolerance response (ATR) has been shown to enhance sessile cells resistance to strong acid challenge (Chorianopoulos, Giaouris, Grigoraki, Skandamis, & Nychas, 2011). Moreover, comparable counts to the ones shown in Fig. 1D were obtained for *L. monocytogenes* EGD-e WT when co-cultured with 7 industrial *Lactobacillus* spp. isolates in BHI-Mn-G for 48 h and 72 h at 30 °C (Fig. S2). Most probably the combination of the protective effect of *Lactobacillus* spp. biofilm biomass and *L. monocytogenes* ATR development, and induction of other general stress response mechanisms results in better survival in biofilms. Therefore, the role of the general stress response regulator *sigB* was investigated in more detail in single and mixed species biofilm formation.

3.2. Role of *sigB* in *L. monocytogenes* EGD-e single and mixed species biofilm formation

The deletion of *sigB* in *L. monocytogenes* EGD-e significantly reduced single species biofilm formation ($P < 0.05$) in BHI-Mn-G (Fig. 2A) and also in BHI confirming previous studies (van der Veen & Abee, 2010) and in BHI-G at 20 °C (Fig. S3). The importance of *sigB* in single species biofilm formation in BHI was also demonstrated by Schwab et al. (2005). However, no reduction in BHI-Mn was observed; counts for the *sigB* mutant were not significantly different ($P > 0.05$) at both time points from those obtained for the WT (shown in Fig. 1A). This pointed again to a role of manganese in stress management of *L. monocytogenes*. In contrast, the role of *sigB* was not significant in all four tested media at 30 °C highlighting that when *L. monocytogenes* forms biofilms faster the role of *sigB* (Fig. 2A, S3) seems to be less important.

In mixed species biofilms, the deletion of *sigB* in *L. monocytogenes* EGD-e did not affect the biofilm formation in BHI, BHI-Mn and BHI-G both at 20 °C and 30 °C (data not shown). The same trend was obtained by Schwab et al. (2005) when studying *L. monocytogenes* mixed species biofilm performance in co-culture with *Pseudomonas fluorescences* in BHI concluding that *sigB* was not essential for mixed species biofilm development. However, in our study when using BHI-Mn-G a significantly reduced performance of *L. monocytogenes* $\Delta sigB$ was observed in mixed species biofilms at 20 °C after 24 h (1 log₁₀ cfu/ml reduction) and 48 h (almost 2 log₁₀ cfu/ml reduction) and at 30 °C after 48 h (>1 log₁₀ cfu/ml reduction) ($P < 0.05$) (Fig. 2B), which might be explained by the reduced tolerance to enhanced acid stress of the mutant at these low pH conditions when co-cultured with *L. plantarum*. It is conceivable that SigB-controlled intracellular and extracellular glutamate decarboxylase (GAD) systems (Cotter & Hill, 2003; Feehily et al., 2014; Karatzas, Suur, & O'Byrne, 2012) and/or the arginine deiminase system (Chen et al., 2011; Ryan, Begley, Gahan, & Hill, 2009) contribute to survival in these conditions due to their capacity to facilitate intracellular pH homeostasis by consuming protons in a decarboxylation reaction and production of ammonium ions, respectively (Gahan & Hill, 2014).

3.3. *L. monocytogenes* biofilm formation in glucose-rich medium with the addition of salt

L. monocytogenes EGD-e WT, the *sigB* mutant, its complementary counterpart and *L. plantarum* WCFS1 were pre-cultured in BHI-Mn-G and BHI-Mn-G with 3% salt (BHI-Mn-G-salt), after which single and mixed species biofilm formation of *L. monocytogenes* EGD-e in co-culture with *L. plantarum* WCFS1 was investigated. This allowed us to

quantify biofilm levels when salt was added only during pre-culturing but not during biofilm formation (−+), when salt was only added during biofilm formation (−+), and when salt was added both during pre-culturing and biofilm formation (++). The addition of 3% NaCl during pre-culturing and single species biofilm formation had no effect on *L. monocytogenes* WT counts and the *sigB* mutant counts at 20 °C or 30 °C (data not shown). Previous studies showed that concentrations between 2% and 4% salt increased *L. monocytogenes* biofilm biomass based on CV bioassay (Jensen, Larsen, Ingmer, Vogel, & Gram, 2007; Pan, Breidt, & Gorski, 2010), but since CV binds non-specifically to culturable and dead cells and to biofilm matrix components, this does not necessarily imply differences in cell counts as we have shown in our study. Significant differences were also not observed by Chorianopoulos et al. (2011) at 16 °C in sessile biofilms produced by *L. monocytogenes* Scott A.

However, in mixed-species biofilms the addition of 3% salt to BHI-Mn-G during pre-culturing and/or biofilm formation protected *L. monocytogenes* WT at 30 °C (Fig. 3), but not at 20 °C (data not shown), whereas the addition of salt did not affect the *L. plantarum* counts at both temperatures. At 30 °C acidification was faster and reached pH 3.6 at 24 h and the addition of salt provided protection, which showed to be *sigB*-dependent (Fig. 3). This might point to a possible cross-protective effect of salt-stress tolerance towards progressive acidification. Indeed, salt stress was shown to enhance acid stress robustness of *L. monocytogenes* (Skandamis, Yoon, Stopforth, Kendall, & Sofos, 2008). SigB-dependence may be explained by activation of its regulon members including osmolyte transporters (Mujahid, Orsi, Vangay, Boor, & Wiedmann, 2013) that mediate intracellular accumulation of protective compatible solutes such as betaine and/or carnitine present in BHI, that contribute to protein (enzyme) and membrane stability under a range of stress conditions (Abee & Wouters, 1999; Gahan & Hill, 2014). Alternatively, there might be a role for the previously described SigB-controlled five-gene stress survival islet (SSI-1) containing *pva*, implicated in bile tolerance, and *gadD1* and *gadT1*. This stress survival islet contributes to the growth of *L. monocytogenes* in suboptimal acidic and/or osmotic conditions (Ryan, Begley, Hill, & Gahan, 2010) and is also present in EGD-e (Hein et al., 2011).

4. Conclusions

Our study showed that glucose-rich medium reduced *L. monocytogenes* single and mixed species biofilm formation with *L. plantarum* both at 20 °C and 30 °C. Addition of MnSO₄ restored biofilm performance of *L. monocytogenes* around the pH growth limit, denoting to an important role of manganese in *L. monocytogenes* stress management.

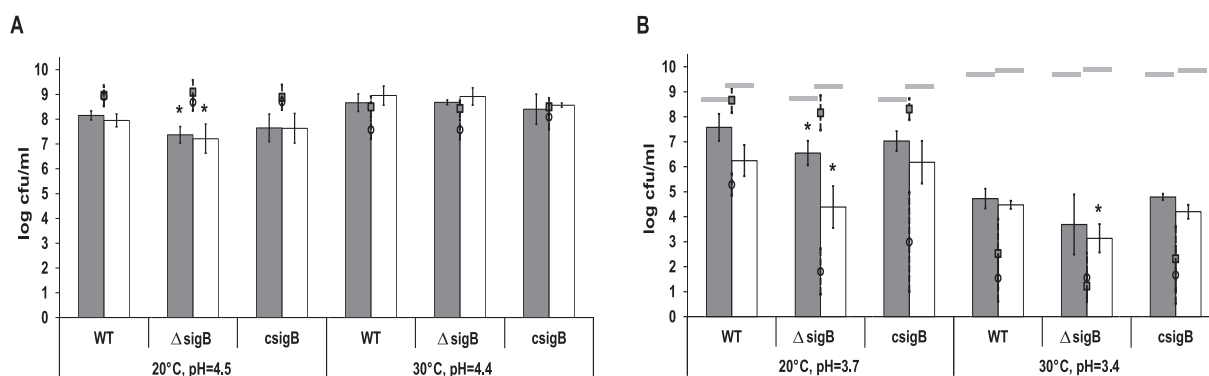


Fig. 2. Comparative *L. monocytogenes* EGD-e biofilm formation and supernatant counts between the wild-type strain, $\Delta sigB$ strain and the complementary mutant (*csigB*) mono-cultured (A) and co-cultured (B) with *L. plantarum* WCFS1 at 20 °C and 30 °C in 12-well polystyrene plates in BHI-Mn-G. The graphs show the average biofilm counts (log₁₀ cfu/ml, bars after 24 h (grey) and 48 h (white)) and *L. monocytogenes* supernatant counts (log₁₀ cfu/ml, symbols after 24 h (square) and 48 h (circle)) of 3 biologically independent experiments; *L. plantarum* WCFS1 average biofilm counts (log₁₀ cfu/ml) in the mixed biofilms (B: light grey lines after 24 h and 48 h, supernatant counts not shown). *, biofilm formation significantly different from the wild type strain. pH measured after 48 h, no significant differences in pH values between WT, $\Delta sigB$ and *csigB*.

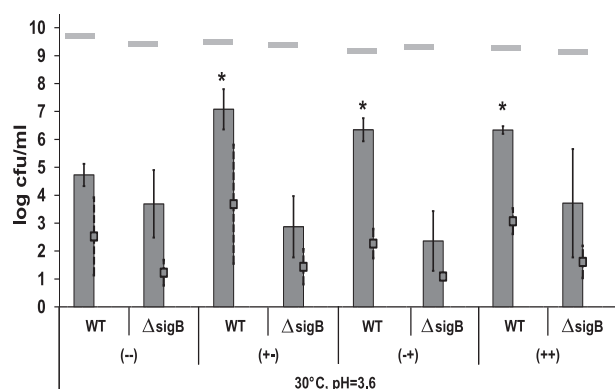


Fig. 3. *L. monocytogenes* EGD-e biofilm formation and supernatant counts co-cultured with *L. plantarum* WCFS1 at 30 °C in 12-well polystyrene plates in BHI-Mn-G without addition of 3% salt during pre-culturing and biofilm formation (---), pre-cultured with the addition of 3% of salt and not during biofilm formation (+-), salt added only during biofilm formation (-+) or added both during pre-culturing and biofilm formation (++). The graph shows the average *L. monocytogenes* EGD-e biofilm (bars) and supernatant (square symbols) counts of 3 biologically independent experiments after 24 h; *L. plantarum* WCFS1 average biofilm counts (log₁₀ cfu/ml) in the mixed biofilms (light grey lines, supernatant counts not shown). *; biofilm formation significantly different from the wild type strain cultured without salt (---). No significant differences in pH values between WT and ΔsigB for all conditions.

However, the contribution of *L. monocytogenes* in biofilms co-cultured with *L. plantarum* was remarkably low in glucose-rich medium and especially at 30 °C, most probably due to the severe acidic conditions promoted by *L. plantarum* homolactic fermentation. Survival in these acidic mixed species biofilms appeared to be sigB-dependent, and the addition of salt protected *L. monocytogenes*. Taken all together, *L. monocytogenes* demonstrated to be a robust survivor in biofilm growth mode, which could contribute to food contamination risks.

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